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Influence of salinity on embryogenesis, survival, growth and oxygen consumption in embryos and yolk-sac larvae of the Nile tilapia

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ABSTRACT

The Nile tilapia is a euryhaline species that offers considerable potential for culture in low-salinity waters. In order to investigate the ability of this species to be reared during the hatchery stages in brackish water, the effects of varying low salinities (0, 7.5, 15, 20 and 25 ppt) on hatchability, survival, growth and energetic parameters until yolk-sac absorption were assessed. Salinity up to 20 ppt was tolerable, although reduced hatching rates at 15 and 20 ppt suggest that these salinities may be less than optimal. Optimum timing of transfer of eggs from freshwater to elevated salinities was 3–4 h post-fertilisation, following manual stripping and fertilisation of eggs, however increasing incubation salinity lengthened the time taken to hatch. Dry body weight was related to salinity, with larvae in salinities greater than 15 ppt displaying, at hatch, a significantly (GLM: $\rho<0.05$) lower body weight but containing greater yolk reserves than those in freshwater or lower salinities. Survival at yolk-sac absorption displayed a significant (GLM; $\rho<0.05$) inverse relationship with increasing salinity and mortalities were particularly heavy in the higher salinities of 15, 20 and 25 ppt. Mortalities occurred primarily during early yolk-sac development yet stabilised from day 5 post-hatch onwards. Salinity had a negative effect on yolk absorption efficiency (YAE). Salinity-related differences in oxygen consumption rates were not detectable until day 3 post-hatch. Oxygen consumption rates of larvae in freshwater between 3 and 6 days post-hatch were always significantly higher (GLM: $\rho<0.05$) than those in 7.5, 15, 20 and 25 ppt, however, on day 9 post-hatch this pattern was reversed and freshwater larvae had a significantly lower QO_2 than those in elevated salinities. Salinity had a significant inverse effect on larval standard length, with elevated salinities producing shorter larvae from hatch until day 6 post-hatch, after which time there was no significant differences between treatments. Salinity had a significant effect on whole larval dry weight, with heavier larvae in elevated salinities throughout the yolk-sac period (GLM; $\rho<0.05$).

1. Introduction

Tilapiine fishes, despite being freshwater species, display an ability to tolerate a broad range of naturally occurring variations in environmental salinities (Philippart and Ruwet, 1982). Amongst the tilapia species of aquacultural interest, there is a clearly defined species specificity of tolerance; many species are broadly euryhaline whilst others are restricted to fresh or low-salinity water. Numerous reviews of salinity tolerance for various cultured tilapias have been published *i.e.* Balerin and Hatton, 1979; Chervinski, 1982; El-Sayed, 2006; Hickling, 1963; Kirk, 1972; Perschbacher, 1992; Prunet and Bournancin, 1989; Stickney, 1986; Suresh and Lin, 1992. The Nile tilapia (*Oreochromis niloticus*), which has now extended well beyond its natural range, dominates tilapia aquaculture because of its adaptability and fast growth rate (Macintosh and Little, 1995; Shelton, 2002). Although *O. niloticus* is not considered to be amongst the most tolerant of the cultured tilapia species, it still offers considerable potential for culture in low-salinity water (Stickney, 1986; Suresh and Lin, 1992).

For many years it has been recognised that the culture of euryhaline fish species in brackish water or marine systems could potentially provide animal protein in areas where freshwater resources are limited (Loya and Fishelson, 1969) and land that is marginal for agriculture. In recent times, the rapidly increasing competition for freshwater for urbanisation, industrialisation and agricultural activities has limited the scope of freshwater aquaculture, especially in arid regions. Therefore there exists an urgent need to manage marine and brackish water environments more efficiently and to diversify aquacultural practices, either by the introduction of new candidate species or by the adaptation of culture methods for existing species. In response to these pressures, total worldwide aquaculture production in brackish water has risen from 1 564 456 MT in 1997 or 5.47% of total world aquaculture to 3 082 261 MT in 2008 or 7% of total aquaculture production (FAO; FishStat Plus, 2010).

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It is well established that salinity exerts a strong influence on development and growth in early life stages of fishes. The aim of this study was to investigate the interaction between low salinity and performance of eggs and yolk-sac larvae of the Nile tilapia in an attempt to define limits in order to optimise hatchery procedures for culture of this species in low salinity water in areas where freshwater resources are limited. For this purpose freshwater spawned eggs were incubated and yolk-sac larvae reared at a range of salinities (0 to 32 ppt) and hatchability, survival, growth and energetic parameters were evaluated.

2. Materials and methods

2.1. Egg supply and artificial incubation system

In all experiments, eggs were obtained from long-term cultured Nile tilapia stocks held at the Tropical Aquarium, Institute of Aquaculture, University of Stirling, UK. Broodstock were maintained individually in 50 l freshwater aquaria with recirculation at 27 ± 1 °C and fed on artificial pellets (#5 trout pellet, Trouw Aquaculture Limited, Skretting, UK). The light regime was maintained at a 12:12 hour light:dark period. Eggs were obtained by manually stripping eggs from a ripe female into a Petrie dish followed by the addition of freshly stripped milt. After 1–2 min, water was added and eggs placed in their respective incubation unit and designated as a batch.

Independent test incubation units consisted of 20 l plastic aquaria, each with an individual Eheim pump (Series 94051) with 6×1 l plastic bottles with a down-welling system (Rana, 1985). Temperature in the incubation units was maintained at 28 °C ± 1 with individual 300 W thermostatically controlled heaters (Visi-therm, Aquarium-systems, Mentor, Ohio, U.S.A.). The experimental salinities were prepared using conditioned water (tap water aerated and heated to 28 ± 1 °C for 24 h prior to use) and commercial salt (Tropic Marin, Aquarientechnic, D-36367, Germany) and salinity was measured using a salinity refractometer (Instant Ocean Hydrometer, Marineland Labs.) accurate to 1 ppt. About 10% of water was replaced daily in the incubation aquaria to compensate for evaporation and salinity was adjusted accordingly. Dead eggs and larvae were removed daily to prevent fungal infection. The light regime was maintained as for broodstock.

2.2. The effect of salinity on embryo viability

In the first experiment, eggs from a freshly stripped batch were fertilised with freshly stripped milt and were exposed to elevated salinities (0, 7.5, 15, 20, 25 and 32 ppt) according to two transfer regimes; Group A eggs were exposed to experimental salinities immediately following fertilisation, and Group B eggs were exposed to freshwater immediately following fertilisation and incubated for 4 h before being transferred to experimental salinities. Three batches of eggs were used for each group with three replicates per batch and 40 eggs per replicate. Embryo viability *i.e.* following normal development was determined by examination of embryos under a dissecting microscope at 4 and 9 h post-fertilisation for Group A or at 9 h post-fertilisation for Group B.

2.3. The effects of salinity on embryogenesis and hatching success

The effects of elevated salinities on the developmental rate of embryos, hatching success and partitioned *i.e.* body compartment and yolk-sac dry weight of larvae at hatch were investigated in the second experiment. Eggs from a newly fertilised batch were placed in the freshwater incubation system, allowed to develop to the 8–16 cell blastula stage (*i.e.* 3–4 h post-fertilisation) and healthy, normally developing embryos were chosen and randomly allocated to each experimental treatment (0, 7.5, 15, 20 and 25 ppt) with three replicates per treatment and 40 embryos per replicate. Thereafter, normally developing embryos were then taken from the freshwater system at 24 h post-fertilisation (approximately one third epiboly) and at 48 h post-fertilisation (beginning of segmentation period) and transferred, as above, to the experimental treatments. A total of three batches were used.

Developmental rates of embryos *i.e.* time to acquisition of selected ontogenetic characteristics, time until hatching (>50% hatch and 100% hatching) and embryonic survival patterns were noted. In addition, 10 newly hatched larvae were removed randomly from each replicate ($n = 30$), killed in MS222 (tricaine methane sulphonate) and immediately rinsed in distilled water. Each yolk-sac larvae was then dissected using a binocular microscope; the yolk was separated from the larval body and yolk-sac epithelium *i.e.* body compartment, and both were placed separately on pre-weighed foils and oven dried at 60 °C for 24 h, followed by 3 h desiccation and then weighed to the nearest 0.0001 g on an Oxford G2105D balance.

2.4. The effect of salinity on survival and growth rate from hatch to yolk-sac absorption

The third experiment was designed to study the effect of salinity on larval performance from hatch until complete yolk-sac absorption. Embryos were allowed to develop to the 8–16 cell blastula stage in the freshwater incubation system and normally developing embryos were randomly allocated to each experimental treatment (0, 7.5, 15, 20 and 25 ppt) with three replicates per treatment and 40 embryos per replicate. A total of three batches were used. Survival was monitored daily until complete yolk-sac absorption occurred. Further eggs from each batch were also transferred the 8–16 cell blastula stage to an additional three replicate bottles per treatment for growth measurements; a total of 30 larvae per treatment were randomly removed at hatch and subsequently on days 3, 6, 9 post-hatch, killed in MS222 (tricaine methane sulphonate) and rinsed in distilled water. Half of the sample was allocated for larval whole weight whilst the other half was dissected from their yolk and the resulting body compartment. Samples were oven dried at 60 °C for 24 h followed by 3 h desiccation and weighed to the nearest 0.0001 g on an Oxford G2105D balance. YAE (%) was calculated.

2.5. The effect of salinity on metabolic rate

The fourth experiment was designed to study the effects of salinity on the metabolic rate of yolk-sac larvae from hatch to yolk-sac absorption. A Strathkelvin microcathode electrode (Model SI130) attached to a Strathkelvin dissolved oxygen meter (Model 782) was used to measure oxygen consumption of individual yolk-sac larvae. A Strathkelvin glass respiration chamber (Model RC300) with a volume of 3 ml was maintained at a constant temperature by pumping water from a temperature controlled water bath through the glass jacket surrounding the chamber. The chamber was provided with a gently moving stir bar to ensure adequate mixing of the water and a screen was placed above the stir bar to protect the larvae. Calibration to zero and 100% saturation for each salinity was assumed to be equivalent to oxygen levels calculated according to the formulae described in Forstner and Gnaiger (1983). Controls were run for each experimental salinity (*i.e.* treatment water and no larvae) due to an observed decline in the amount of oxygen in the chamber not due to the metabolism of the larvae and the value of this blank was subtracted from the respective respiration values. Initial trials were run to see the effects of stress on oxygen consumption. Transfer of larvae to the respiration chamber appeared to increase O_2 consumption rates which were seen to level out after 5 min within the chamber. Therefore two respiration chambers were used alternatively - an individual larva was placed in the spare respiration chamber 5 min prior to measurement of O_2 consumption so as to allow larvae to acclimatise.

Eggs from three separate batches were allowed to develop to 8–16 cell stage in the freshwater incubation system and normally developing embryos were then randomly allocated to each experimental treatment (0, 7.5, 15 , 20 and 25 ppt). At hatch, 3, 6 and 9 days post-hatch, larvae were removed (n = 9 per salinity) and oxygen consumption rates for individual larvae were measured (μl O_2 indv.$^{-1}$ h^{-1}). Larvae were killed in MS222 (tricaine methane sulphonate), rinsed in distilled water and standard length measurements were carried out, according to May (1971). Individuals were immediately frozen at −70 °C and subsequently oven dried, desiccated and weighed to the nearest 0.0001 g on an Oxford G21050 balance. Data was expressed as QO_2 *i.e.* μl O_2 mg dry weight^{-1} h^{-1}.

2.6. Performance indices

Body compartment gain (BCG) was calculated as follows:

$$\mathrm{BCG(dry\ weight)(mg)} = \text{body compartment at yolk–sac absorption} - \text{body compartment at hatch}.$$

Yolk consumed (YC) was calculated as follows:

$$\mathrm{YC(dry\ weight)(mg)} = \text{yolk at hatch (dry weight)} - \text{yolk at absorption(dry weight)}$$

Yolk absorption efficiency was calculated using the following formula:

$$\mathrm{YAE(\%)} = (\text{body compartment gain (dry weight)} * 100)/ \times \text{yolk consumed during yolk absorption period(dry weight)}.$$

2.7. Statistical analyses

Statistical analyses were carried out using Minitab 16 software using a General Linear Model or one-way analysis of variance (ANOVA) with Tukey's post-hoc pair-wise comparisons. Homogeneity of variance was tested using Levene's test and normality was tested using an Anderson-Darling test. All percentage data were normalised by arcsine square transformation prior to statistical analyses to homogenise the variation and data are presented as back-transformed mean and upper and lower 95% confidence limits.

3. Results

3.1. The effect of salinity on embryo viability

Embryo viability after 9 h was significantly affected ($\rho<0.05$) by salinity, regardless of the timing of post-spawning exposure to treatment salinities; embryos incubated at elevated salinities always displayed a lower viability than those incubated in freshwater (Table 1). Embryonic development was severely inhibited at 32 ppt with no development observed after 9 h, regardless of transfer time. The timing of exposure of eggs to elevated salinities had a significant effect ($\rho<0.05$) on embryo viability after 9 h, with immediate exposure to an elevated salinity (Group A) negatively affecting viability compared with exposure after 4 h (Group B). However, the effects of immediate exposure of eggs to experimental salinities were more apparent after 9 h than after 4 h, with embryos after 4 h displaying a significantly reduced viability ($\rho<0.05$) only for salinities of 20 ppt and above, whereas embryos after 9 h displayed a significantly reduced viability ($\rho<0.05$) for all salinities (Table 1).

3.2. The effects of salinity on embryonic development, hatching success and dry weights of yolk-sac larvae at hatch

Nile tilapia embryos developed and hatched in all salinities tested (0, 7.5, 15, 20 and 25 ppt), however hatching rates, regardless of transfer time, was always significantly reduced ($\rho<0.05$) in elevated salinities as compared to freshwater. Acclimation regime *i.e.* timing of transfer, similarly had an effect on hatching rate. Eggs transferred from freshwater to elevated salinities either at 24 h post-fertilisation or at 48 h post-fertilisation displayed a lower hatching rate, compared with those transferred at the 3–4 hour stage (Fig. 1).

Embryonic survival curves for eggs transferred to experimental salinities are shown in Fig. 2. Survival was inversely related, regardless of timing of transfer, to salinity. Mortalities for embryos transferred at 3–4 h post-fertilisation occurred immediately after transfer to the higher experimental salinities and showed a gradual decline in survival thereafter until hatch. Mortality of embryos transferred at a later stage *i.e.* 24 h and 48 h post-fertilisation similarly occurred rapidly within the first 24 h following transfer, and then remained constant until hatch. For embryos transferred at 48 h post-fertilisation, there was an increase in mortality at around 82 h post-fertilisation, especially at elevated salinities *i.e.* 20 and 25 ppt.

Table 1
Effects of salinity on viability (%) of Nile tilapia embryos according to transfer time to experimental salinities. Statistical analyses, means and 95% confidence limits were calculated on arcsine square transformed data of three trials with three replicates per trial (n = 40 eggs per replicate). Values in the same column sharing a common superscript are not significantly different (P<0.05); asterisks next to values for 9 hours post-spawning sampling in Group B denote a significant difference between corresponding value in Group A.

Sampling point (hours post-fertilisation)	Egg viability (%): mean and 95% confidence limits (upper – lower) 4	9
Group A: eggs transferred to treatments immediately after spawning [a]		
0 ppt	94.5 (96–92)[a]	94.1 (96–91)[a]
7.5 ppt	92.9 (94–91)[a]	65.4 (67–63)[b]
15 ppt	94.1 (96–91)[a]	73.2 (74–71)[c]
20 ppt	85.7 (87–84)[b]	55.6 (57–53)[d]
25 ppt	62.8 (64–60)[c]	22.7 (23–21)[e]
32 ppt [#]	6.7 (8–4)	0
Group B: eggs transferred to treatments after 4 hours incubation in freshwater [b]		
0 ppt	95.6 (97–92)	99.2 (99–97)[a]
7.5 ppt	-	97.4 (98–96)[b]*
15 ppt	-	85.3 (86–84)[c]*
20 ppt	-	86.8 (87–86)[c]*
25 ppt	-	56.7 (57–55)[d]*
32 ppt [#]	-	0

[a] Group A initial egg weight (mg) = 3.2 ± 0.02; [b] Group B initial egg weight (mg) = 3.6 ± 0.03. [#] Data excluded from statistical analyses.

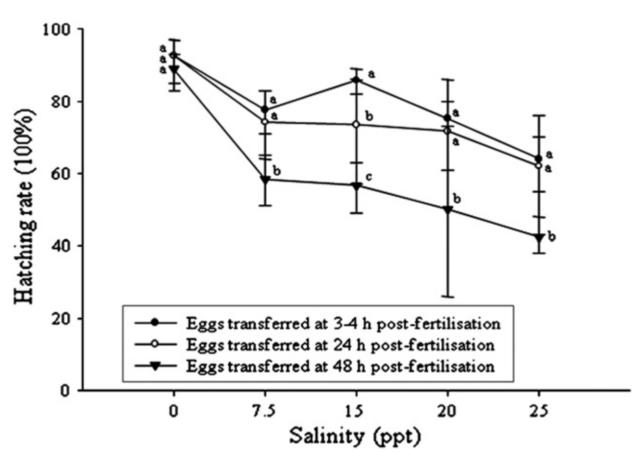

Fig. 1. Comparison of hatching rates of Nile tilapia eggs subjected to varying post-fertilisation acclimation regimes. Mean and 95% confidence limits were calculated on arcsine square transformed data of three batches with three replicates per batch (n = 40 eggs per replicate). Different letters indicate significant differences between acclimation regimes ($\rho<0.05$).

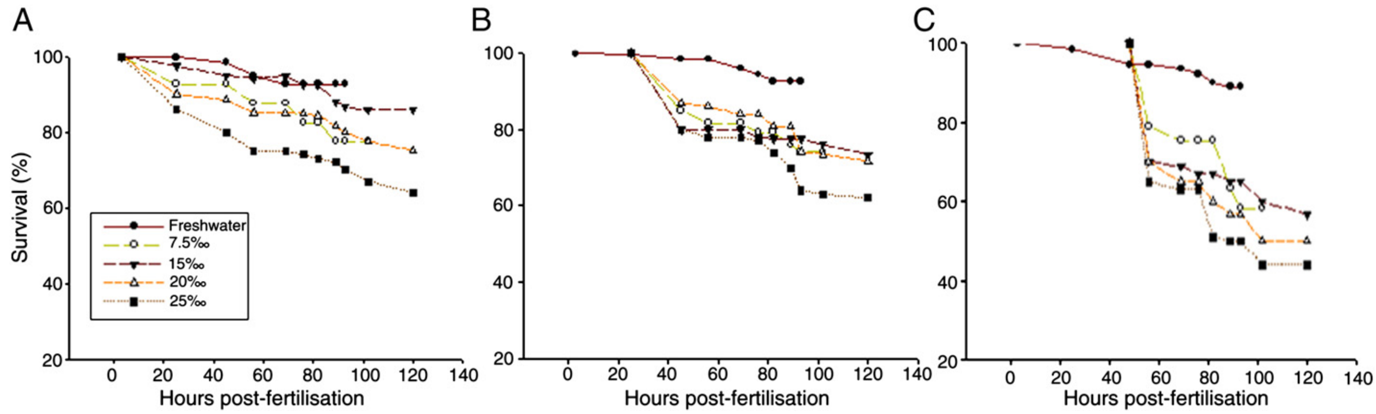

Fig. 2. Survival curves for embryos transferred to various experimental salinities. Means were calculated on arcsine square transformed data of three batches with three replicates per batch (n = 40 eggs per replicate). A) Embryos transferred at 3–4 h post-fertilisation, B) Embryos transferred at 24 h post-fertilisation and C) Embryos transferred at 48 h post-fertilisation.

Increasing incubation salinity lengthened the time taken to reach selected embryonic stages (Fig. 3). Time to 100% hatch for embryos transferred at 3–4 h post-fertilisation was inversely related to incubation salinity with hatching times ranging from 120 h for embryos incubated in 15, 20 and 25 ppt to 93 hours for embryos incubated in freshwater.

Body compartment weight and yolk weight were inversely related *i.e.* embryos incubated in 20 ppt produced yolk-sac larvae at hatching with a lower body weight but containing greater yolk reserves ($\rho<0.05$) (Fig. 4).

3.3. The effect of salinity on growth rate and survival of yolk-sac larvae from hatch to yolk sac absorption

Larval survival at complete yolk-sac absorption displayed a significant inverse relationship with increasing salinity at the salinities tested (Table 2). Survival curves of larvae up to yolk-sac absorption for the three trials are shown in Fig. 5. Mortality occurred in all treatments, primarily during early development *i.e.* from hatch to 5 days post-hatch. Mortalities increased with increasing salinity and were particularly heavy in the higher salinities of 15, 20 and 25 ppt. Following initial mortality, survival generally stabilised from 5 d post-hatch. The survival pattern for yolk-sac larvae reared in 7.5 ppt was similar to that observed for yolk-sac larvae reared in freshwater.

The mean dry body compartment weight and whole dry weight for larvae at hatch and at yolk-sac absorption are shown in Table 2. At hatch, yolk-sac larvae had a greater whole dry body weight in elevated salinities (>15 ppt) compared with those in freshwater. Similarly, at yolk-sac absorption, yolk-sac larvae incubated and reared in elevated salinities (>15 ppt) had a higher whole body weight than those in freshwater. Yolk-sac absorption efficiency (YAE) was salinity dependant; larvae reared in 20 and 25 ppt showed a lower YAE than those reared in freshwater, 7.5 and 15 ppt (Table 2).

3.4. The effect of salinity on metabolism of yolk-sac larvae

Salinity-related differences in oxygen consumption rates were not detectable until 3 days post-hatch and, thereafter, mean QO_2 rates increased during development for all salinities. The oxygen consumption rates of larvae in freshwater between days 3–6 post-hatch were always significantly higher ($\rho<0.05$) than those in 7.5, 15, 20 and 25 ppt however on day 9 post-hatch this pattern was reversed and freshwater larvae had a significantly lower QO_2 than those in elevated salinities (Fig. 6).

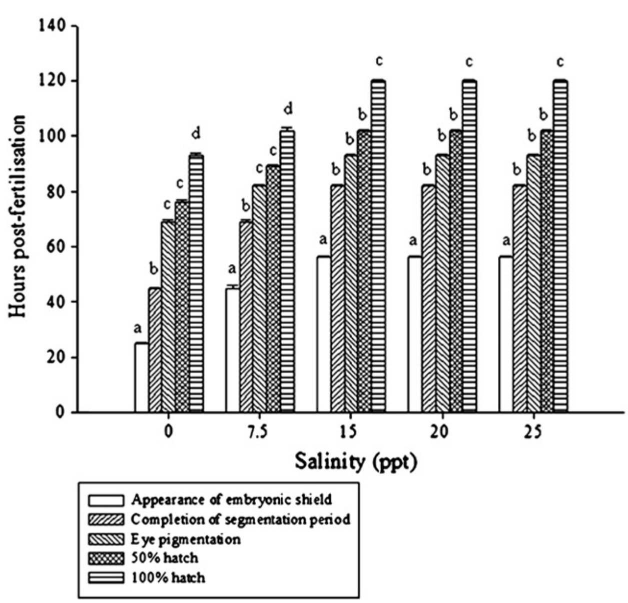

Fig. 3. Effect of incubation salinity on the developmental rate of Nile tilapia embryos transferred to experimental salinities at 3–4 h post-fertilisation. Data points are mean ± S.E. of three batches with three replicates per batch (n = 40 eggs per replicate).). Different letters indicate significant differences between developmental stages ($\rho<0.05$).

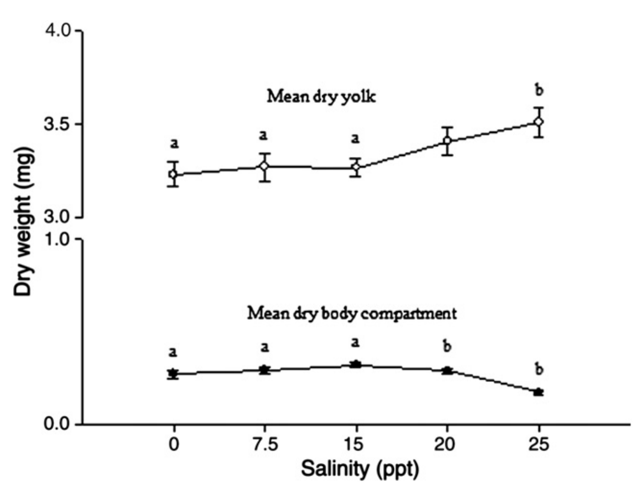

Fig. 4. Effect of incubation salinity on mean dry body compartment (total weight minus yolk-sac) and mean dry yolk weight of newly hatched Nile tilapia larvae. Embryos were transferred 3–4 hours post-fertilisation. Data points are mean ± S.E. of three batches with three replicates per batch (n = 40 eggs per replicate). Different letters denote significant differences between treatments ($\rho<0.05$).

Table 2
Influence of salinity on growth characteristics of Nile tilapia fry from hatch to yolk-sac absorption. Different superscripts indicate statistical differences between treatments (ρ<0.05).

Treatment	Weight at hatch (mg)		Weight at yolk-sac absorption (mg)	Yolk-sac absorption efficiency (%) [1]	Survival (%): mean and 95% confidence limits (upper–lower)
	Whole	Body compartment			
0 ppt	3.9 ± 0.06^{ab}	0.2 ± 0.09^{a}	2.8 ± 0.11^{ab}	61.21	$80.4\ (96–56)^{a}$
7.5 ppt	3.8 ± 0.02^{a}	0.3 ± 0.04^{a}	2.7 ± 0.03^{b}	64.72	$80.2\ (99–39)^{a}$
15 ppt	4.0 ± 0.01^{ab}	0.5 ± 0.10^{b}	3.1 ± 0.07^{a}	64.57	$66.7\ (75–58)^{a}$
20 ppt	4.2 ± 0.17^{b}	0.4 ± 0.03^{b}	3.1 ± 0.12^{a}	59.75	$68.5\ (86–46)^{a}$
25 ppt	4.1 ± 0.02^{b}	0.3 ± 0.02^{b}	3.1 ± 0.09^{a}	57.66	$68.0\ (90–47)^{a}$

[1] Yolk Absorption Efficiency, YAE (%) = ((body compartment gain (dry weight) - yolk consumed during yolk absorption period (dry weight)) x 100).Values are means ± S.E. of three replicates per Trial (n = 30 yolk-sac larvae per replicate). Statistical analyses, mean and 95% confidence limits were calculated on arcsine square transformed data for survival data (%).

Salinity had a significant effect (ρ<0.05) on larval dry weight with heavier larvae in 15 ppt and above from hatch until day 6 post-hatch. Salinity similarly had a significant effect (ρ<0.05) on larval standard length with shorter larvae in 20 and 25 ppt except on day 9 post-hatch where there was no significant difference between treatments (Table 3).

4. Discussion

4.1. Effects of salinity on embryogenesis

The results presented in this study indicate that freshly fertilised eggs from freshwater maintained parents transferred immediately to test salinities of 7.5, 15, 20 and 25 ppt displayed significantly lower fertilisation rates after 9 h than those transferred after 4 h prior incubation in freshwater. Alderdice (1988; p. 237) suggests that, at spawning, eggs face 'the first major regulatory challenge' as they are subjected to any major changes in the osmotic and ionic properties of the spawning water. Prior to this, they are subject to homeostatic regulation by the adult regulatory system, with contacts between oocyte and follicular cell microvilli allowing transfer of nutrients and ions. Post-ovulation but pre-spawning, their plasma membrane appears to be relatively permeable to water and will reflect the corresponding osmolality of the ovarian fluid, a level approximately one third that of ambient water (Holliday and Jones, 1967). After spawning and activation of the egg following fertilisation, cortical alveolar exocytosis causes imbibition of water from the external environment across the chorion, forming the perivitelline space. Immediately following this, regulation and maintenance of the integrity of the egg appears to be achieved by the resistive maintenance of a tight plasma membrane and limited trans-membrane water and ion fluxes (Kao and Chambers, 1954), making the egg 'rather impermeant' (Bennett et al., 1981). This theory of post-spawned egg impermeability is supported by Swanson (1996) who transferred eggs of milkfish (*Chanos chanos*), spawned at 32–36 ppt to lower and higher salinities at the cleavage-blastula stage (*i.e.* 2–5 h post-fertilisation) and observed no swelling or shrinkage of the eggs in response to osmotic gradients. Therefore, it is suggested that the ionic and osmotic composition of the environment immediately post-spawning is vital to egg viability and that if newly spawned eggs and the external salinity were of different osmoconcentrations, the resulting imbalance could affect fertilisation and egg viability, as is seen in this study.

There is some evidence to support this hypothesis in the Nile tilapia. Watanabe and Kuo (1985) looked at the effects of spawning salinity upon *O. niloticus* eggs and found mean hatching success for eggs spawned and incubated at 10 ppt (32.7%) and 15 ppt (36.9%) to be similar to those spawned and incubated in freshwater (30.9%), By direct comparison, in another study, Watanabe et al. (1985) found a reduced hatching success of *O. niloticus* eggs spawned in freshwater but directly transferred one day post-spawning to varying salinities (0, 5, 10, 15, 20, 25 and 32 ppt).

In this study Nile tilapia embryos were able to tolerate salinity challenge across the range of 7.5 to 25 ppt, with results indicating that salinity had a significant negative effect on hatching rates. Mortality was 100% in the elevated salinity of 32 ppt and indeed failure of Nile tilapia embryos to develop at this salinity has previously been reported (Watanabe and Kuo, 1985; Watanabe et al., 1985). In general, there is a paucity of work on the effects of salinity on the embryogenesis of teleosts that mainly focuses on euryhaline marine

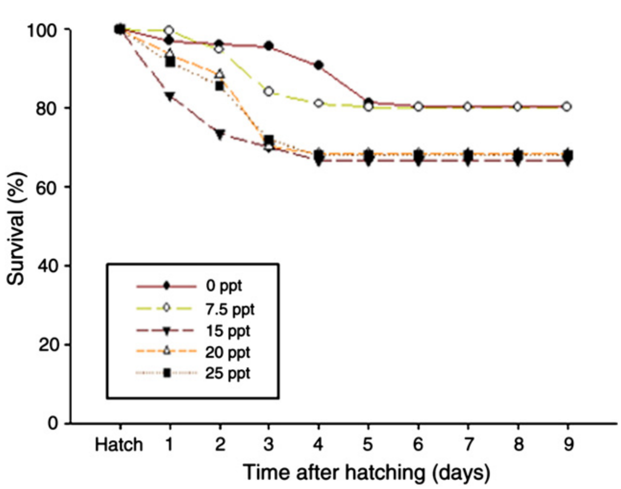

Fig. 5. Survival curves for Nile tilapia larvae reared at various experimental salinities following transfer at 3–4 h post-fertilisation. Means were calculated on arcsine square transformed data of three batches with three replicates per batch (n = 30 yolk-sac larvae per replicate).

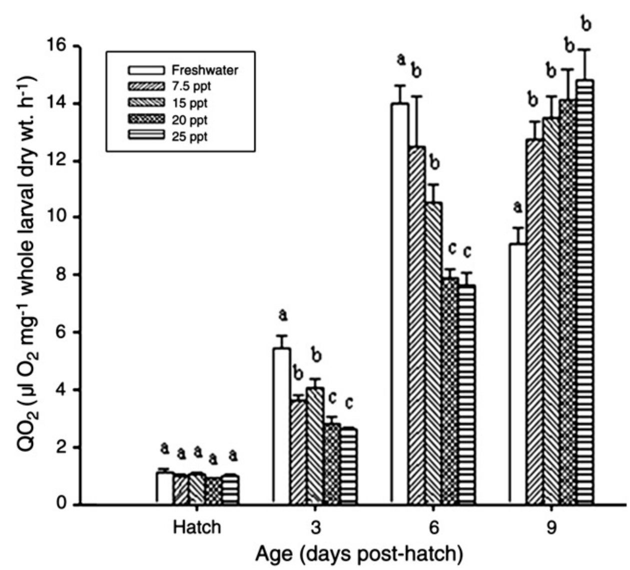

Fig. 6. Effects of salinity on oxygen consumption of Nile tilapia yolk-sac larvae. Oxygen consumption is expressed as QO_2 (µl O_2 mg^{-1} whole larval dry wt. h^{-1}). Values represent mean ± S.E. of data from three Trials (n = 9 yolk-sac larvae per Trial). Different letters above each bar indicate significant differences amongst treatments (ρ<0.05).

Table 3
Effect of salinity on larval standard length (mm) and larval dry weight (mg). Values represent mean ± S.E. Different superscripts (a,b,c,d) and (w,x,y,z) indicate significant differences between treatments and between days (ρ<0.05), respectively.

Salinity:	0 ppt	7.5 ppt	15 ppt	20 ppt	25 ppt
Standard length (mm)					
Hatch	$5.4 \pm 0.08\ ^{a}{}_{w}$	$5.3 \pm 0.04\ ^{a}{}_{w}$	$5.3 \pm 0.08\ ^{a}{}_{w}$	$4.9 \pm 0.08\ ^{b}{}_{w}$	$4.6 \pm 0.06\ ^{b}{}_{w}$
3 dph	$7.1 \pm 0.05\ ^{a}{}_{x}$	$6.5 \pm 0.07\ ^{b}{}_{x}$	$6.8 \pm 0.12\ ^{ab}{}_{x}$	$6.0 \pm 0.08\ ^{c}{}_{x}$	$6.1 \pm 0.79\ ^{c}{}_{x}$
6 dph	$7.8 \pm 0.08\ ^{a}{}_{y}$	$7.7 \pm 0.11\ ^{a}{}_{y}$	$7.7 \pm 0.08\ ^{a}{}_{y}$	$7.3 \pm 0.07\ ^{b}{}_{y}$	$7.4 \pm 0.07\ ^{b}{}_{y}$
9 dp	$8.0 \pm 0.04\ ^{a}{}_{y}$	$8.1 \pm 0.11\ ^{a}{}_{y}$	$8.1 \pm 0.13\ ^{a}{}_{y}$	$8.2 \pm 0.06\ ^{a}{}_{z}$	$8.2 \pm 0.05\ ^{a}{}_{z}$
Dry weight (mg)					
Hatch	$2.8 \pm 0.13\ ^{a}{}_{y}$	$3.0 \pm 0.15\ ^{b}{}_{x}$	$3.4 \pm 0.13\ ^{b}{}_{z}$	$3.5 \pm 0.15\ ^{b}{}_{y}$	$3.6 \pm 0.14\ ^{b}{}_{y}$
3 dph	$2.4 \pm 0.07\ ^{a}{}_{x}$	$2.8 \pm 0.20\ ^{b}{}_{x}$	$3.1 \pm 0.12\ ^{b}{}_{y}$	$3.6 \pm 0.11\ ^{b}{}_{y}$	$3.6 \pm 0.12\ ^{b}{}_{y}$
6 dph	$2.0 \pm 0.13\ ^{a}{}_{w}$	$2.4 \pm 0.24\ ^{a}{}_{w}$	$2.6 \pm 0.10\ ^{a}{}_{x}$	$3.3 \pm 0.14\ ^{b}{}_{x}$	$3.4 \pm 0.11\ ^{b}{}_{x}$
9 dph	$2.4 \pm 0.16\ ^{a}{}_{x}$	$2.4 \pm 0.09\ ^{a}{}_{w}$	$2.3 \pm 0.11\ ^{a}{}_{w}$	$2.7 \pm 0.11\ ^{b}{}_{w}$	$2.8 \pm 0.14\ ^{b}{}_{w}$

species; many marine species spawn offshore, relying on offshore currents to disperse their progeny, which often brings eggs and larvae to coastal areas displaying varying salinities. Studies on marine species generally show that hatching rates of eggs are adversely affected as salinity moves from the normal salinity range encountered in nature *e.g.* eggs of the milkfish (*C. chanos*) spawned in 32–36 ppt but transferred 2–5 h post-fertilisation to varying salinities showed a reduced hatching success (50% hatch) in both lower (15 and 20 ppt) and higher (50 and 55 ppt salinities (Swanson, 1996), and eggs of the mullet (*Mugil cephalus*) transferred at the gastrula stage from a spawning salinity of 30 ppt showed an optimum salinity range for hatching of 30 to 40 ppt and reduced hatching rates in lower (10–25 ppt) and higher (45–50 ppt) salinities (Lee and Menu, 1981).

The ability of Nile tilapia to tolerate changes in salinity during embryogenesis in the present study was, likewise, clearly influenced by the stage of embryonic development upon transfer. The results obtained indicate that there was a significant increase in hatching rate of Nile tilapia eggs transferred at 3–4 h post-fertilisation compared with embryos transferred at a later stage *i.e.* 24 or 48 h post-fertilisation. The pattern of embryonic survival seemed to follow the same trend, with mortalities increasing rapidly following transfer, this being especially pronounced in those eggs transferred at later stages of embryonic development (48 h). Conversely, Lee and Menu (1981) reported that eggs of the grey mullet (*M. cephalus*) transferred from the salinity of spawning (30 ppt) at the late gastrula stage (approx. 12 h post-spawning) showed a wider range in tolerance *i.e.* 20–45 ppt than eggs transferred at the 2-blastomere stage (approx. 1 h post-spawning) to the test salinities where the best hatching was reported at the reduced salinity of 35 ppt. Similarly, Lee et al. (1981) reported that fertilised eggs of the euryhaline Northern whiting (*Sillago sihama*) were more tolerant to salinity change at later stages of development.

Alderdice (1988) describes the establishment of osmotic regulation during embryogenesis as beginning during gastrulation and being in place by yolk-plug closure or completion of epiboly. The appearance of integumental chloride cells during gastrulation marks the start of the selective restriction of ions transfer or active ionoregulation (Guggino, 1980a, 1980b). The first appearance of chloride cells on the yolk-sac epithelium of dechorionated Mozambique tilapia (*Oreochromis mossambicus*) embryos has been reported at 26 h post-fertilization but only at 48 h post-fertilisation has the presence of apical crypts indicated functionality (Lin et al., 1999). Ayson et al. (1994) likewise observed chloride cells on the yolk-sac epithelium of *O. mossambicus* embryos at 30 h post-fertilization (*c.* 2 h after the beginning of the gastrula stage) in both fresh and seawater, distributed underneath the pavement cells, with functional apical openings noted at 48 h post-fertilization or half-way to hatching. It would be expected, therefore, that if ontogenetic changes in osmoregulatory function *i.e.* appearance of chloride cells, confer adaptability during gastrulation, eggs should be more tolerant to transfer to elevated salinities as embryogenesis progresses. However, this is different to what has been reported in the present work. It is suggested here that the increased incidence of mortality immediately post-transfer at 48 h and the subsequently significantly reduced hatching rate as compared to those transferred at 3–4 h post-fertilisation seen in this study, supports the theory that the increase in permeability of the chorion during gastrulation allows passage of the external media *i.e.* both water and salts, into the developing egg and puts an osmoregulatory strain on an embryo that is not yet able to cope, as chloride cells are only just beginning to gain full functionality.

4.2. Effects of salinity on survival and growth of yolk-sac larvae

In this study, mortalities occurred in all salinities during the first few days after hatching, declining by day 3 post-hatch and then levelling out by day 5 post-hatch until yolk-sac absorption. Mortality was especially pronounced at higher salinities. This suggests that Nile tilapia face the greatest osmoregulatory challenge immediately after hatching, yet show an increasing capacity to maintain ionic and osmotic balance that is conferred ontogenically through the yolk-sac period. This is different to previous studies concerning the ability of larvae of marine species to withstand abrupt salinity challenge *e.g.* Young and Dueñas (1993) reported that 12 h post-hatch rabbitfish larvae (*Siganus guttatus*) could tolerate transfer to salinity within the range of 10–45 ppt yet at 24 h post-hatch to the reduced salinity range of 14–37 ppt. Correspondingly, Banks et al. (1991) reported that 1 day post-hatch spotted sea trout larvae (*Cynoscion nebulosus*) could tolerate salinity ranges of 4–40 ppt and at 3 days post-hatch could tolerate 8–32 ppt. Similarly, Estudillo et al. (2000) reported a longer LT_{50} for newly hatched larvae of the red snapper (*Lutjanus argentimaculatus*) than for larvae of 7, 14 or 21 days post-hatch when abruptly transferred from 32 ppt to a lower salinity.

It is well documented that teleost yolk-sac larvae are able to maintain osmotic and ionic gradients between their internal and external environments (Alderdice, 1988; Guggino, 1980a, 1980b; Kaneko et al., 1995), due mainly to the presence of numerous extrabranchial chloride cells commonly observed on the abdominal epithelium of the yolk-sac and other body surfaces of fish larvae. Integumental chloride cells have been reported in the post-embryonic stages of several teleost species (see review Varsamos et al., 2005). A distinct spatial shift in ionocyte distribution from body surface to branchial areas during ontogeny is acknowledged and has been reported in several species *i.e.* the European sea bass (*Dicentrachus labrax*) (Varsamos et al., 2002), the killifish (*Fundulus hereroclitus*) (Katoh et al., 2000), the Mozambique tilapia (*O. mossambicus*) (van der Heijden et al., 1999; Yanagie et al., 2009) and the Nile tilapia (*O. niloticus*) (Fridman et al., 2011). The presence of chloride cells on branchial epithelia during yolk-sac stages has been confirmed by Li et al. (1995) who identified them on branchial tissue of developing larvae of freshwater Mozambique tilapia (*O. mossambicus*) at 3 days post-hatch before lamellae were fully formed, showing a 50% increase in density of chloride cells by 10 days post-hatch (*i.e.* at yolk-sac absorption) with this density remaining constant up to the adult stage. Correspondingly, Hiroi et al. (1998) described the first appearance of branchial chloride cells on the gill filaments before differentiation of the lamellae at 8 days post-hatch in the Japanese flounder (*Paralichthys olivaceus*). Therefore it is suggested that the temporal patterns of survival following hatching that were observed in this study may indicate that, with the development of the branchial system and the increase in numbers of chloride cells by 3 days post-hatch onwards, the larvae are better able to cope with osmoregulatory challenge.

4.3. Effects of salinity on metabolism of yolk-sac larvae

Oxygen consumption has been used as an indirect indicator of metabolism in fishes (Cech, 1990) and consumption rates, in response to variations in environmental salinities, have been employed in an attempt to assess the energetic costs of osmoregulation in a wide range of teleost species. Variations in results have often led to confusion (Swanson, 1998). The assumptions that metabolic rates are lowest at iso-osmotic salinities because of minimal osmoregulatory costs and that the extra oxygen consumed in increasingly non iso-osmotic media is proportional to the increase in osmoregulatory requirements can be supported by studies carried out in juvenile and adult teleosts *e.g.* Nile tilapia (*O. niloticus*) (Farmer and Beamish, 1969), rainbow trout (*Oncorhynchus mykiss*) (Rao, 1968), sea bream (*Sparus sarba*) (Woo and Kelly, 1995) and *O. mossambicus* x *Oreochromis. hornorum* hybrids (Febry and Lutz, 1987). However, contrary results have been reported by Morgan and Iwama (1991) in the juvenile rainbow trout (*O. mykiss*) and Chinook salmon (*Oncorhynchus tshawytscha*), these authors describing lowest oxygen consumption in freshwater with increasing consumption in increasing salinity and by Job (1969 a and b) in *O. mossambicus* who described higher oxygen uptake in 12.5 ppt than in either fresh water or 100% seawater. Ron et al. (1995) also described a significantly ($\rho<0.05$) lower consumption in 20 month old *O. mossambicus* reared in seawater rather than in freshwater and Iwama et al. (1997) similarly demonstrated lower oxygen consumption rates in *O. mossambicus* acclimated to sea water as compared to fresh water and hyper-saline water (1.6 x sea water).

Varying results have also been reported for teleost eggs and larval stages. No salinity-related differences in oxygen consumption rates have been observed in eggs and larvae of the Pacific sardine (*Sardinops sagax*) (Lasker and Theilacker, 1962), eggs and larvae of the herring (*Clupea harengus*) (Holliday et al., 1964), eggs of the grubby (*Myoxocephalus aenaeus*) and longhorn sculpin (*Myoxocephalus octodecemspinosus*) (Walsh and Lund, 1989), eggs and larvae of the striped mullet (*M. cephalus*) (Walsh et al., 1991), eggs and yolk-sac larvae of the milkfish (*C. chanos*) (Walsh et al., 1991), eggs and alevins of Rainbow trout (*O. mykiss*), Chinook salmon (*O. tshawytsha*) (Morgan et al., 1992) and hatch to 35 day old fry in the Nile tilapia (*O. niloticus*) (De Silva et al., 1986). However, in accordance with this study, Swanson (1996) reported a significant effect ($\rho<0.05$) of salinity on oxygen consumption rates of milkfish (*C. chanos*) eggs and yolk-sac larvae. In general these discrepancies may be due to limitations in accurately estimating osmoregulatory costs that reflect both methodological and physiological factors (Swanson, 1998), including a lack of standardisation in methods or systems of oxygen measurement as well as inconsistency of acclimation regimes often resulting in the confounding effect of stress, these effect combining with variations in age, size, ecology and natural distribution of species investigated.

In the present study, weight-specific oxygen consumption rates ($\mu l\ O_2$ mg. dry wt.$^{-1}$ h^{-1}) (QO_2) were seen to increase during the yolk-sac period for all salinities tested. Metabolic rates are strongly influenced by developmental stage and by the amount of metabolically active tissue (Swanson, 1996) and indeed, linear relationships of oxygen consumption with age during embryogenesis have already been demonstrated for eggs and newly hatched larvae of milkfish (*C. chanos*) (Swanson, 1996), eggs and yolk-sac larvae of milkfish (*C. chanos*) (Walsh et al., 1991), eggs and larvae of striped mullet (*M. cephalus*) (Walsh et al., 1989), early life stages of the common carp (*Cyprinus carpio*) (Kaushik et al., 1982). On the other hand, non-linear relationships have been reported for Atlantic halibut eggs (*Hippoglossus hippoglossus*) (Finn et al., 1991), yolk-sac larvae of large-mouth bass (*Micropterus salmoides*) (Laurence, 1969), eggs and larvae of cod (*Gadus morhua*) (Davenport and Lonning, 1980) and yolk-sac larvae of Randall's rabbitfish (*S. randalli*) (Collins and Nelson, 1993). However, the limitations in methods of estimation that exist for variations in oxygen consumption rates as a result of salinity as described above may similarly apply here.

A significant effect of salinity was observed for weight-specific oxygen consumption rates ($\mu l\ O_2$ mg. dry wt.$^{-1}$ h^{-1}) (QO_2) from 3 days post-hatch onwards. However, salinity related variations in QO_2 do not appear to reflect a direct metabolic cost of osmoregulation, as differences were not related to the magnitude of osmotic gradient between larvae and surrounding water. This is in agreement with observations in milkfish eggs (*C. chanos*) transferred from a spawning salinity of 32–36 ppt to either an iso-osmotic range of 15–20 ppt or to a strongly hyper-osmotic range of 50–55 ppt where equally low oxygen consumption rates were measured for both ranges (Swanson, 1996). Nevertheless, salinity during early life stages may indirectly influence larval development rate and hence activity levels and resulting energetic cost (Swanson, 1996). It has been suggested that muscular activity increases the metabolic rate of yolk-sac larvae by mixing the perivitelline fluid which in turn facilitates gas exchange (Peterson and Martin-Robichaud, 1983). Salinity-related differences observed in this study from 3 days post-hatch onwards support this suggestion, since they occur only once larval movement has commenced, as thereafter it could be seen that salinity clearly had an effect on rate of yolk absorption and growth between 3 and 6 days post-hatch. Whilst fish in freshwater had a lower mean dry weight than those in elevated salinities, they had a greater standard length, indicating that more yolk-sac had been absorbed and used for somatic growth. The linear relationships between dry weight and QO_2 and standard length and O_2 demonstrated in this study shows that both lighter fish and longer fish have greater oxygen demand.

The reduction of the diffusive capacity of the epithelia and the resulting dependency on branchial respiration as larvae develop (Kamler, 1992) could explain why the more developed larvae in freshwater were more active and showed a higher metabolic rate, this being supported by branchial respiration. Depressed activity of larvae with larger yolk-reserves at higher salinities could account for the significantly lower QO_2 value for 7.5, 15 and 20 ppt compared with freshwater on day 3 and day 6. Tsuzuki et al. (2008) reported that larvae of the silversides (*Odontesthes hatcheri* and *Odontesthes bonariensis)* were visibly less active at 30 ppt than at lower salinities. De Silva et al. (1986) similarly demonstrated that larval activity may be responsible for increasing metabolic consumption; their study of un-anaesthetised fresh water *O. niloticus* larvae showed a 3-fold increase in oxygen consumption from 3.4 to 10.09 $\mu l\ O_2$ indv.$^{-1}$ h^{-1} between 0 and 14 h post-hatch and 2–3 days post-hatch, yet basal oxygen consumption measured for anaesthetised larvae did not show such a large change, increasing from 2.55 to 3.06 $\mu l\ O_2$ indv.$^{-1}$ h^{-1}.

4.4. Application and perspectives in Nile tilapia culture

Limited information is available on salinity tolerance during early life stages of Nile tilapia, a species that currently dominates freshwater tilapia culture, and, in addition offers potential as a candidate species for brackish water aquaculture. This work confirms the euryhaline nature of the early life stages of the Nile tilapia, showing that salinities up to 20 ppt were tolerable, although reduced hatching rates and survival to yolk-sac absorption at 15 and 20 ppt suggests that these salinities may be less than optimal. Optimum timing of transfer of eggs from freshwater to elevated salinities is 3–4 h post-fertilisation, following manual stripping and fertilisation of eggs. This finding has direct practical applications in hatcheries where, in general, spawning occurs naturally and eggs are removed from the buccal cavity of the females and transferred to elevated salinities several days after spawning has occurred. Early low salinity exposure would not only minimise freshwater hatchery requirements but also may confer a pre-adaptation before transfer to higher salinities for on-growing. Therefore the results of this study have significant practical implications for both the development of hatchery production methods and for the future potential for aquaculture of this species in brackish water.

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